
INSTALL PROCESS AND PROCEDURE 2025 / 2026

1.0 ONT Standards

The brains of the Customer's Network; this delivers the services to the other devices

The correct steps are to always deal with the ONT first, in any install/swap/troubleshooting.

The ONT is always the primary device.

Fiber Cleaning:

Before plugging into the ONT ensure you perform Fiber Cleaning using

- 2.5mm fiber click cleaner (for SC/APC and SC/UPC)
- Lint-free fiber wipes or fiber cleaning cards
- Fiber-safe cleaning solution (99% isopropyl alcohol or approved solvent)
- Fiber scope or visual inspection tool (if available)
 - Remove Dust Cap
 - Inspect the Connector (if scope available)
 - Use Dry Click Cleaner
 - Insert the connector into the cleaner
 - Push straight in until it clicks once
 - Do not double-click without rotating the cleaner head
 - If Still Dirty – Use Wet-to-Dry Method
 - Cleaning Bulkhead Ports or OLT Adapters
 - Use a 2.5mm Bulkhead Click Cleaner
 - Insert the cleaner into the adapter or port
 - Click once to clean the ferrule inside

Install ONT/GPON

- Power up, connect fiber, download and upgrade firmware
 - Do not circumvent this by rebooting the device
 - The process can take 7 to 20 minutes
 - This will ensure ONT is provisioned and ready for other devices to be provisioned and to come online
 - Do not add devices in OFS until the ONT has locked in and is stable
 - Add ONT first in OFS and then other devices
 - If the ONT is Adtran verify that the ONT is not a recalled 622

ONT Validation

- Upon completion of work the tech will provide a light reading
 - The light reading will be recorded on the light reading line of the Install Health Check
 - A photograph of the tech's meter readings will be included on all completed orders
 - **IMPORTANT: Light readings will be collected at both the NID and the ONT moving forward.**
 - **ONT light reading should be between -14 and no more than -19.5dB**
 - **There should NOT be greater than a -1dB loss between the NID light reading and the ONT light reading. If there is, troubleshoot the fiber- in; including connectors**
- ***This will assist in determining the integrity of the fiber in***

2.0 Router Standards

The EVO 1840AP: WI-FI 6E AP Router is the preferred router for all installs.

The router is the next step in the customer installation process

Router Installation

Onboarding the router: **[MODIFIED]**

- The Evolution Wi-Fi 6E router must be added to the customer account through OFS or FSC if assistance is required.
- The use of the HomePass app **was** the primary method for adding the router to the Plume management portal. **This is no longer the case.** as this piece is now automated.
- Auto Provisioning will create the customer's portal automatically once the technician adds the EVO router to the account through OFS. **Same applies for extenders.** No additional app setup, downloads, or customer walkthroughs are required for portal creation.
- Technicians must still confirm that: The router was successfully added in OFS, the device shows online and healthy internet connectivity is active before adding any TiVo+ equipment. Frontline validation and speed tests are completed as usual.

This updated process ensures faster activation, reduces extra steps for techs, and prevents duplicate or mismatched portal setups. The Home Pass will be used to educate the customer on usage at the very end (**if applicable**).

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Note: There will be a slight delay while the portal builds and adds the router to the portal. Once it's built, the tech can log into Frontline to set the SSID & PSWD.

EVO1840AP: WI-FI 6E AP ROUTER / MESH BEACON

EVO1840AP Wi-Fi 6E dual-operational AP router and mesh beacon in the 2+4+2 tri-band antenna configuration enables service providers to safely provide high-quality Wi-Fi in the subscriber's home. By using Wi-Fi6E technology, customers enjoy the fastest Wi-Fi technology available in the marketplace. Service providers can leverage improved remote management and monitoring capabilities to reduce support costs.
Plume OS 3.2 certification completed May 30, 2023.

Technical Specifications	
Wi-Fi Technology	- Tri-band (2.4GHz, 5GHz, 6GHz) [2.4G] 2x2 11ax [5G] 4x4 11ax [6G] 2x2 11ax - RAM: 1 GB DDR3 - NAND: 512MB - BCM6756 (Main), BCM6715 (Wi-Fi)
Key Features	- Evolution Mesh, EasyMesh and OpenSync Under development - IPv4/IPv6 dual stack support 204080/160 MHz channels 802.11 a/b/g/n/ac/ax 1 x 2.5GbE WAN, 3 x 1GbE LAN - BLE 5.0
Ease of Use	Plume OS 3.2 supports HomePass, Haystack, WorkPass, Uprise, Harvest (Certification complete May 30, 2023). - Entering firmware validation for custom OoKla URL and advanced BLE features (static IP, PPPoE, VLAN) shortly. Expect GA September.





Wi-Fi 6E
(2.4GHz)



Tri-Band
Concurrent



Mobile App
Support

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Technicians will assist the customer with downloading and setting up the HomePass app and provide customer education on the app's features.

- **NOTE: Onboarding the router must be completed prior to leaving the customer premises. If the customer is unable to complete the onboarding through email verification the technician can complete in the Frontline portal under "Tech Dashboard"**
- Select "Not Complete" to complete the onboarding process for the customer.
- Whenever possible, the tech should look to provide a wired solution for the customer
- Ideally the router is located in a central location within the premise
- The same process is to be followed for extender installation.
- Router / Extender Validation
 - Upon completion of work the tech will provide an Overall Health Score reading from Frontline
 - Router / Extender speed test from Frontline
 - A photograph of the tech's Frontline readings will be included on all completed orders

Wi-Fi Setup (Connect all customer Wi-Fi devices before leaving)

- Technicians must use the plume Frontline Tier 1 application to determine the signal strength to each device
- Ensure there are no parallel networks in the home (especially true if an upgrade)
- Validate all devices connected to the router via the RSSI reading determined from the Topology within Frontline

- **Ensure that the ACCT# in OFS matches in Frontline, what you see on the Tech Dashboard page.**
- Wi-Fi Validation is obtained via Frontline using the Topology feature and overlaying the RSSI scores.
- All readings of devices at the time of install to be included in Install Health Check

RSSI Score Range

RANGE		IMPACT
-30 dBm	-45 dBm	Max theoretical achievable strength and may be too close.
-46 dBm	-55 dBm	Ideal real-world signal strength. Great for high demand applications such as 4K streaming and transferring large files.
-56 dBm	-65 dBm	Minimum for reliable packet delivery. Good for streaming and most applications.
-66 dBm	-75 dBm	Throughput will likely be compromised for high bandwidth applications. IoT devices should still operate normally. Small changes in the environment may result in poor performance.
-76 dBm	-99 dBm	Minimum for basic connectivity. Packet delivery will be unreliable and functionality will be impacted. Streaming will likely buffer.

3.0 SBBTV Set-Top Installation

TiVo Force 1 STBs

- **Install & Activate TiVo Boxes**
 - Validate customer account has the appropriate SBB TV Rate Codes
 - **If it does not, do not install** Force 1 STBs on to the customer's account
 - Work with the FSC to have the proper rates codes added to the customer's account.
 - ***DO NOT install TiVo+ STB until you've confirmed that the internet is working first. (NO EXCEPTIONS!).***
 - Connect TiVo boxes to the router via Ethernet (**the preferred method whenever available**) or MoCA.
 - Power on each TiVo box and follow on-screen activation prompts.

- Confirm channel lineup, VOD access, and TiVo+ apps functionality.
- Validate all IP STBs (regardless of type) RSSI Score in the Frontline portal
- Prioritize video service for each Force 1 device
- Pair remotes and demonstrate operation to the customer.

- **Close-Out Tasks**

- Record all STB RSSI scores on the Install Health Checklist
- Review the installation with the customer, answer questions.
- Upload all compliance photos/screenshots to the ticket via OFS.
- Record serial numbers of ONT, router, and TiVo boxes in the work order.
- Close the trouble ticket or work order with detailed notes & capture customer signature.

- Quick Reminder:

- | • Term • | Meaning | • Used For |
|----------|-----------------------|---------------------------------------|
| • dB • | Ratio (gain/loss) | • Amplifier gain, signal attenuation |
| • dBm • | Absolute signal level | • Power level at ONT, RF input levels |

Summary

1. The ONT is first
2. The router is next and should be hardwired and centrally located whenever possible
3. Wait for the internet to come up and work before adding the TiVo+ STB to the account
4. The same router rules for any extenders apply
5. All devices at the time of installation need to be recorded and an RSSI score should be shown
6. When installing SBBTV, validate the proper rate codes are on the account prior to adding to the customer's account
7. Ensure all STBs are given Prioritization (to Video) on the customer's network
8. All facets of the install should be recorded in OFS under the Install Health Check
9. All six required pictures of the install should be uploaded at completion

Remember! Follow the order of Operations

ONT – Online and provisioned

Router – Only add to OFS after the above and ensure internet is working

STBs and other CPE devices – Only add to OFS after the ONT is online and provisioned and the router is online and connected to the internet. Ensure you have asked the customer if there is any other device or devices (i.e. cameras, Ring Doorbell) that you can assist with providing network access

4.0 Install Health Checklist (mandatory)

1. ONT Light Score TX_RX @ NID and @ ONT
2. Bip Errors Up_Down
3. Light Level Location
4. Overall Health Score >80 (If below 80, check router placement and set up)
5. Router Speed Download_Upload (within 90% of subscribed speed, if below call FSC)
6. Extender 1 Speed Download_Upload
7. Extender 2 Speed Download_Upload
8. Extender 3 Speed Download_Upload
9. Total Devices Online – At the time of the installation. Ensure you have checked with customer that all devices are installed while you are there.
10. Total Devices <75 Db (less than 75Db)
11. Router Location
12. Speed Package Installed
13. STB RSSI Scores

5.0 Upload Required Pictures (mandatory)

1. ONT Light Levels (via MSI/light meter)
2. Plume Frontline Topology screen w/ RSSI levels
3. Plume Frontline Overall Score
4. Speed test results via PLUME Frontline
5. ONT installed/mounted
6. Router/APs installed/mounted

The above material was covered with me, and I understand the contents

Signature:

Date:
